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Introduction

In today's world, it's hard to imagine life without online communication. We message, share files, and talk to people halfway across the planet within seconds. Social networks, emails, and chat apps have changed the way we connect both in our personal lives and at work. But as communication gets faster and easier, it also becomes riskier. Cyberattacks, data leaks, and hacking incidents are now part of everyday news. Without proper protection, sensitive data like passwords, private messages, or business information can fall into the wrong hands.

To deal with these problems, secure chat applications have become increasingly important. Their main idea is simple to make sure only the sender, and receiver can read a message. This is made possible through end-to-end encryption (E2EE). Even if someone manages to intercept a message, they won't be able to read it without the correct decryption key. This approach keeps conversations private and shields users from hackers, man-in-the-middle attacks, and unauthorized access.

In this project, I developed a secure chat system that uses a mix of proven encryption methods such as RSA, AES, and Diffie-Hellman key exchange. RSA helps with public and private key encryption, AES makes data encryption fast and reliable, and Diffie-Hellman ensures both sides can share keys safely. Together, they form a strong defense against data theft or message tampering.

Ultimately, the goal of this project is to build a chat system that isn't just functional but also trustworthy. By combining cryptography with network programming, it shows how secure and private communication can be achieved in the digital age. Something we all depend on more than ever.

Problem Statement

Nowadays, it's almost impossible to imagine life without online communication. Billions of messages are sent every day through chat apps, emails, and social platforms. This shift has made it incredibly easy to connect with people, both personally and professionally. However, that same convenience also brings new dangers. Cyberattacks like phishing, eavesdropping, and man-in-the-middle (MITM) attacks are happening more often than ever. Recent cybersecurity studies report that more than 80% of data breaches involve unauthorized access to private or organizational communication (Tamer & AL-Sammarraie, 2020). Clearly, there's a growing need for communication systems that are both private and secure.

Many popular chat services still rely on centralized servers to store user messages. While some of them advertise that they use encryption, not all actually implement full end-to-end encryption (E2EE). That means messages could still be accessed by the company itself or by hackers if the servers are compromised (Saranya & Suguna, 2024). For instance, if encryption keys are stored on the same server, a single security breach could reveal every user's conversation. Weak key management and incomplete encryption can easily lead to privacy leaks, identity theft, or even financial loss.

This isn't just a personal issue either. Businesses, schools, and government offices also exchange sensitive data online every day. Without a secure messaging system, they face real risks from espionage to data corruption or even surveillance. That's why developing a communication platform that is secure, efficient, and easy to use has become so important in our connected world.

To tackle these challenges, this project focuses on creating a Secure Chat Application that uses multiple layers of encryption such as RSA, AES, and Diffie-Hellman key exchange. Each algorithm plays a different but essential role: AES provides fast data encryption, RSA secures key exchange, and Diffie-Hellman ensures safe key sharing between users. Research by Dahri et al. (2025), Sharaf and Ahmad (2024), and Yusfrizal and Meizar (2024) confirm that combining these algorithms can offer reliable protection in modern messaging systems. With its focus on end-to-end encryption and secure key exchange, this project aims to show how strong cryptography can make online communication both safer and more trustworthy.

Literature Review

Title	Author	Year	Focus Area (Project Relevance)	Problem Addressed
Implementation of Secure End-to-End Encrypted Chat Application Using Diffie–Hellman Key Exchange and AES-256 in a Microservice Architecture	Dahri et al.	2025	Hybrid Cryptography; DH/AES for E2EE in scalable architectures.	Increasing threats require secure, scalable communication systems that protect data via DH key exchange and AES-256 encryption.
Secure Chat Application with an End-to-End Encryption	Sharaf and Ahmad	2024	Hybrid Cryptography; RSA/AES implementation in real-time chat.	Lack of robust security in chat apps requires a dual-layered system using RSA for key exchange and AES for message encryption.
SECURE CHAT USING ENCRYPTION	Saranya and Suguna	2024	E2EE Implementation; General Cryptography in messaging.	Widespread use of unsecured messaging platforms leads to data breaches and privacy concerns, necessitating mandatory encryption.
Key Management Using Combination of Diffie–Hellman Key Exchange with AES Encryption	Yusfrizal and Meizar	2024	Key Management DH/AES combination.	Vulnerabilities in session key distribution, proposes using Diffie-Hellman to securely generate the AES key.
A Secure Messaging Application with Unbreakable End to End Encryption	Bhatt and Sharma	2021	Protocol Design; E2EE for strong privacy (X3DH, Double Ratchet).	Governments/service providers seek "backdoors" into messaging, necessitating advanced protocols that guarantee data is technically

				inaccessible to third parties.
Securing Text Messages Using E2EE (End-To-End Encryption) on Instant Messaging Applications	Hassan and Al-Sammarraie	2025	Review of E2EE in commercial applications (WhatsApp, Skype).	Need to understand and evaluate the varying E2EE methods and security levels implemented across popular commercial chat applications.
A Review on Hybrid Encryption Techniques: Combining AES and RSA for Enhanced Security	Srivastava et al.	2021	Review of Hybrid Schemes; RSA/AES performance.	Single encryption algorithms fail to balance the speed (symmetric) and key security (asymmetric) required for bulk data transfer.
Secure Data Transmission using Hybrid Cryptography (AES and RSA) for Real-time Application	Babu and Kumar	2021	Performance Analysis; Hybrid AES/RSA in real-time.	Slow key exchange in purely asymmetric systems; hybrid approach aims to achieve both high security (RSA) and fast encryption speed (AES) for real-time data.
A Comparison Study on Key Exchange-Authentication protocol	Al-Dahhan et al.	2018	Key Exchange Comparison (DH vs. others).	Evaluating various key exchange protocols to determine the most efficient and secure method for use in communication systems.
A Study of Cryptography Algorithms Used in Messenger Applications	Singh et al.	2020	Review of Cryptographic Methods (AES, DH in industry).	Analyzing existing messenger applications to identify which cryptographic algorithms are successfully used to provide E2EE.

Research Questions and Objectives

With increasing public concern about online privacy and data security, the demand for encrypted communication systems is also growing. This project aims to develop a secure chat application that ensures the confidentiality, integrity, and authenticity of data through encryption and secure key exchange mechanisms. To guide this research, we have proposed several research questions.

Research Questions

1. How can end-to-end encryption be effectively implemented in chat applications to ensure that only authorized users can access transmitted messages?
2. In real-time communication, which combination of encryption algorithms such as RSA, AES, and Diffie-Hellman achieves the best balance between performance and security?
3. How can the system prevent unauthorized access, such as man-in-the-middle (MITM) attacks or data interception attacks, during message transmission and key exchange?
4. How can usability and efficiency be balanced while maintaining high encryption standards?

Objectives

1. To design and develop a secure chat application using end-to-end encryption with RSA, AES, and Diffie-Hellman algorithms.
2. To ensure message confidentiality, integrity, and authentication through secure key exchange and encryption technologies.
3. To evaluate the application's performance and user experience in terms of encryption speed, message delivery, and ease of use.
4. To demonstrate how modern encryption methods enhance the security of digital communication platforms.

Methodology

This project uses an Agile methodology to design and build a secure chat application. Agile was selected because it encourages flexibility, teamwork, and ongoing feedback, all of which are essential when developing software that involves constant testing, especially for data security and encryption. Compared to traditional methods like the Waterfall model, Agile allows updates and improvements to happen continuously as new ideas or security concerns arise. This iterative cycle helps ensure that the final application is both secure and practical, meeting the main goal of delivering a reliable encrypted communication platform.

The development work is divided into several short phases, or “sprints,” where each sprint focuses on a specific function or improvement. In this project, the Agile process is broken down into five key stages:

1. Requirement Gathering and Planning

The first step was to understand what users need. To do this, we reviewed and compared existing chat applications to identify key requirements such as real-time communication, end-to-end encryption, and secure key exchange. A product backlog was then created, listing all the necessary features, including encryption modules, user interface components, and network functionality. These tasks were prioritized for each sprint based on importance and complexity.

2. System Design

The second stage involves planning the overall system architecture. The design follows either a client-server or peer-to-peer model, developed using Python socket programming or alternatively Java/Node.js. For encryption, AES is used to secure message content, RSA for key encryption, and Diffie-Hellman for key exchange. The user interface focuses on simplicity and usability to ensure smooth communication without sacrificing security.

3. Development and Implementation

Development takes place over several short cycles, each lasting about one to two weeks. The process begins with building the basic chat function through socket programming for sending and receiving messages. Next, AES encryption and decryption are implemented to secure message content. RSA or Diffie-Hellman is then integrated for secure key exchange. After that, the user interface is refined for better usability and

performance. Finally, each cycle ends with debugging, testing, and optimization.

4. Testing and Evaluation

Testing happens continuously throughout the process. Unit tests are used to confirm that encryption, key generation, and messaging functions work correctly. Integration testing ensures that components such as the client, server, and encryption modules operate together smoothly. Security testing includes simulated attacks like packet sniffing or man-in-the-middle attempts to confirm the system's confidentiality and resilience.

5. Deployment and Review

Once testing is complete, the finished application is deployed in a controlled setting for real-world evaluation. User feedback is gathered to identify any weaknesses in usability, performance, or encryption efficiency. Future updates may include adding multi-user chat, biometric authentication, or other advanced features to strengthen both security and user experience.

Tools, Models and Frameworks

- Programming Language: Python and using socket and cryptography libraries
- Encryption Framework: RSA, AES and Diffie-Hellman key exchange
- Development Approach: Agile Methodology
- Testing tools: Wireshark for network monitoring and debugging

By using Agile, the project remains adaptable and user-focused, ensuring that the final Secure Chat Application meets both security standards and usability expectations.

Scope and Limitation

This project focuses on building a secure chat application that allows users to send and receive text messages in real time through either a client-server or peer-to-peer setup. The main goal is to make communication safe by using end-to-end encryption (E2EE), which ensures that only the sender and receiver can read the exchanged messages. To achieve this, the system will use a mix of encryption methods such as RSA, AES, and the Diffie-Hellman key exchange. These techniques together help protect messages during transmission, handle encryption and decryption, and make the key-sharing process more secure.

The application will be created using Python socket programming, allowing users to connect securely and communicate over a local network or the internet. It will include features like message encryption before sending and decryption after receiving, along with protection against common threats such as eavesdropping or man-in-the-middle attacks. For this version, the project will stay focused on text-based communication, so it won't include extras like group chats, multimedia transfers, voice or video calls, or cloud storage.

There are a few limitations to keep in mind. Network delays, differences between devices, or the heavy processing required by encryption might affect the overall performance. The system will mainly be tested in a controlled environment, so the results might not fully reflect how it performs on larger or real-world networks. Still, the project's main purpose is to show how cryptography and programming can work together to create a secure and private way to chat online.

Expected Outcome

The expected outcome of this project is the successful creation of a secure chat application capable of supporting real-time text communication with strong end-to-end encryption (E2EE). The system will demonstrate how encryption methods such as RSA, AES, and Diffie-Hellman key exchange can be combined to build a reliable and protected communication channel. Every message exchanged through the platform is expected to remain confidential, ensuring that only the sender and receiver have access to the actual content.

The final result will be a functional prototype that enables users to send and receive encrypted messages securely over a network. The project will also highlight the significance of secure key exchange and show how encryption helps prevent data interception, tampering, and unauthorized access. Through a series of tests, the system should prove resilient against common cyber threats, including eavesdropping and man-in-the-middle (MITM) attacks.

Beyond the technical aspect, this project aims to enhance understanding of how encryption technologies strengthen modern digital communication. The findings can serve as a useful reference for students, developers, and researchers working in cybersecurity and networking. Ultimately, this study seeks to encourage safer online communication practices and inspire future innovation in lightweight, efficient, and user-friendly secure messaging solutions.

Project Plan

The development of the secure chat application will follow a structured iterative process based on agile methodologies, divided into several key phases: research, design, implementation, testing, and evaluation. Each phase will contribute to building a secure, reliable, and user-friendly communication system capable of real-time encrypted messaging using encryption algorithms.

Phase 1: Research and Requirements Analysis

This phase focuses on identifying user and security requirements. We will research existing chat applications and encryption methods such as RSA, AES, and Diffie-Hellman. The goal is to define system objectives and identify the project's functional and non-functional requirements.

Phase 2: System Design

I will design the system architecture, including client-server or peer-to-peer models, encryption processes, and user interface layout. We will create data flow diagrams and encryption workflow diagrams to visualize the encryption, transmission, and decryption of messages.

Phase 3: Implementation

Development will be conducted in multiple iteration cycles.

Iteration Cycle 1: Develop socket-based chat communication.

Iteration Cycle 2: Integrate AES encryption to ensure message security.

Iteration Cycle 3: Implement RSA or Diffie-Hellman key exchange.

Iteration Cycle 4: Design and refine the graphical user interface.

Phase 4: Testing and Debugging

Unit and integration tests will be conducted to ensure all components function correctly. Security testing will also be performed to verify its ability to resist eavesdropping and unauthorized access.

Phase 5: Evaluation and Documentation

The final prototype will undergo performance, usability, and encryption reliability evaluations. A project report and demonstration will conclude the process.

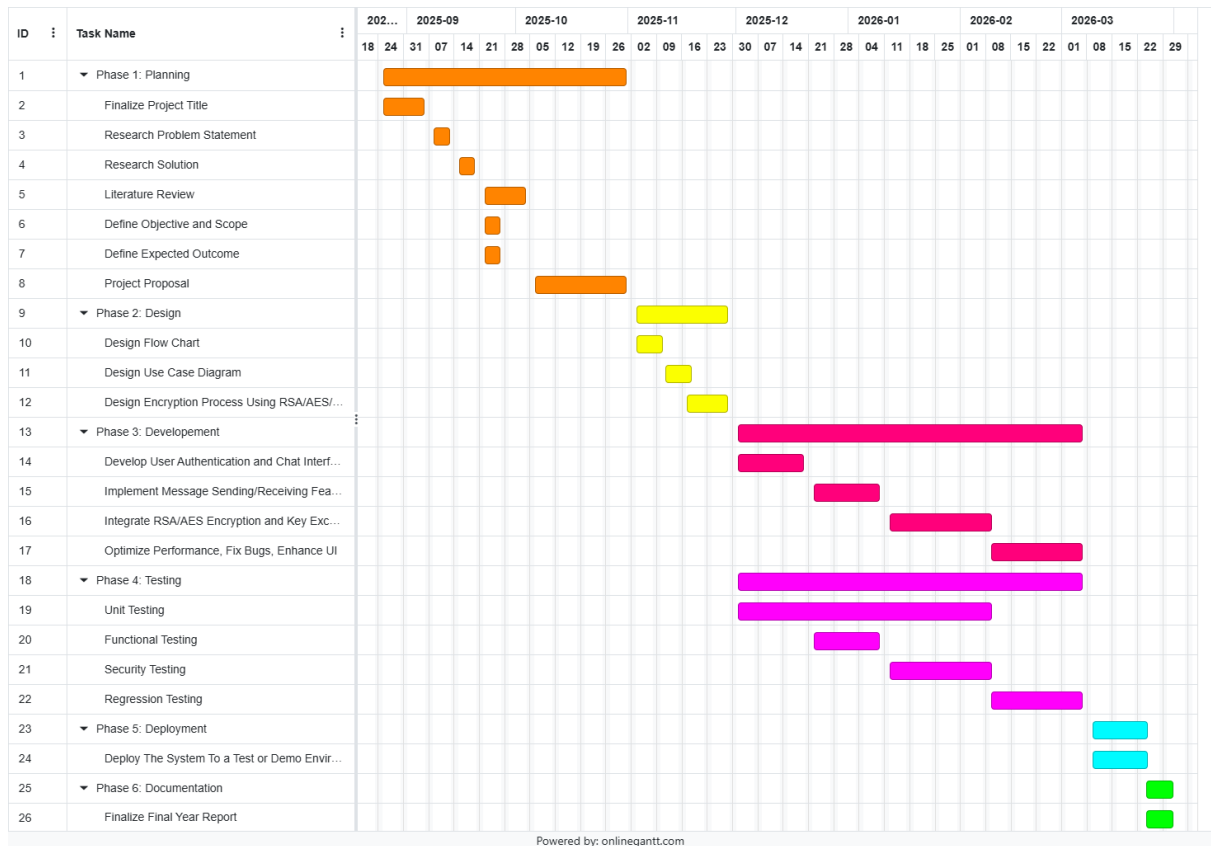


Figure 1

Figure 1 is the Gantt Chart for the project. The start date is from 25 Aug 2025 and end date is on 31 Mar 2026.

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